

# Week 2 Day 2 Hands-on Lab: Storage and Network

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이 lab은 Day 1 PostgreSQL container를 출발점으로 volume, bind mount, custom network, container DNS를 검증한다.

## Phase A: no-volume data loss

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```
docker stop paperclip-pg16 || true
docker rm paperclip-pg16 || true
docker run -d --name paperclip-pg16 -e POSTGRES_PASSWORD=postgres -e
POSTGRES_DB=paperclip -p 15432:5432 postgres:16
```

```
docker exec paperclip-pg16 psql -U postgres -d paperclip -c "\dt"
```

## Phase B: named volume persistence

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```
docker stop paperclip-pg16 || true
docker rm paperclip-pg16 || true
docker volume create paperclip-pg16-data
docker run -d --name paperclip-pg16 -e POSTGRES_PASSWORD=postgres -e
POSTGRES_DB=paperclip -p 15432:5432 -v paperclip-pg16-
data:/var/lib/postgresql/data postgres:16
```

```
docker exec paperclip-pg16 psql -U postgres -d paperclip -c "CREATE TABLE IF
NOT EXISTS notes(id serial PRIMARY KEY, body text); INSERT INTO notes(body)
VALUES ('volume evidence'); SELECT * FROM notes;"
docker stop paperclip-pg16
docker rm paperclip-pg16
docker run -d --name paperclip-pg16-v2 -e POSTGRES_PASSWORD=postgres -e
POSTGRES_DB=paperclip -p 15432:5432 -v paperclip-pg16-
data:/var/lib/postgresql/data postgres:16
docker exec paperclip-pg16-v2 psql -U postgres -d paperclip -c "SELECT * FROM
notes;"
```

## Phase C: volume inspect

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```
docker volume ls | grep paperclip-pg16-data
docker volume inspect paperclip-pg16-data
```

## Phase D: bind mount

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```
mkdir -p week2/day2/tmp-bind/html
printf "<h1>bind mount v1</h1>" > week2/day2/tmp-bind/html/index.html
docker run -d --name paperclip-bind-web -p 18082:80 -v "$PWD/week2/day2/tmp-
bind/html:/usr/share/nginx/html:ro" nginx:alpine
curl -s http://localhost:18082
printf "<h1>bind mount v2</h1>" > week2/day2/tmp-bind/html/index.html
curl -s http://localhost:18082
docker inspect paperclip-bind-web --format "{{ json .Mounts }}"
```

## Phase E: custom network and DNS

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```
docker network create paperclip-day2-net || true
docker run -d --name paperclip-net-pg --network paperclip-day2-net -e
POSTGRES_PASSWORD=postgres -v paperclip-pg16-data:/var/lib/postgresql/data
postgres:16
docker run --rm --network paperclip-day2-net postgres:16 pg_isready -h
paperclip-net-pg -U postgres
```

## Phase F: host port vs container DNS

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```
docker stop paperclip-net-pg || true
docker rm paperclip-net-pg || true
docker run -d --name paperclip-net-pg --network paperclip-day2-net -e
POSTGRES_PASSWORD=postgres -p 15432:5432 -v paperclip-pg16-
data:/var/lib/postgresql/data postgres:16
docker ps --filter name=paperclip-net-pg
docker run --rm --network paperclip-day2-net -e PGPASSWORD=postgres
postgres:16 psql -h paperclip-net-pg -U postgres -d paperclip -c "SELECT 1;"
```

## Cleanup

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```
docker stop paperclip-pg16-v2 paperclip-bind-web paperclip-net-pg || true
docker rm paperclip-pg16-v2 paperclip-bind-web paperclip-net-pg || true
docker network rm paperclip-day2-net || true
rm -rf week2/day2/tmp-bind
# 데이터 삭제가 필요할 때만 실행
# docker volume rm paperclip-pg16-data
```

## Evidence

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- no-volume data loss result
- named volume persistence result
- bind mount v1/v2 result
- custom network DNS result
- cleanup decision